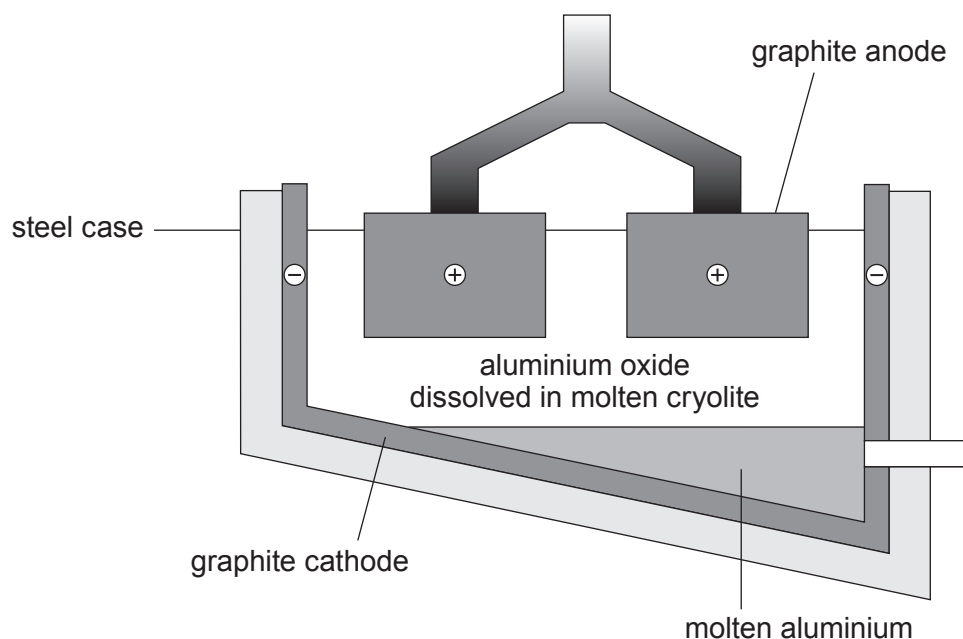


4. (a) Aluminium metal is produced from aluminium oxide in a process called electrolysis. The diagram shows a cell used in electrolysis.



- (i) Underline the correct word in the brackets to complete each sentence. [4]

Aluminium oxide is a (**compound** / mixture / element) of aluminium and oxygen.

Cryolite is added to (**raise** / lower / maintain) the temperature used for electrolysis.

Electrolysis uses a lot of (**electrical** / solar / chemical) energy.

Aluminium is produced as a (**solid** / liquid / gas).



- (ii) Aluminium oxide contains the ions Al^{3+} and O^{2-} .

Give the formula of aluminium oxide.

[1]

.....

- (iii) Iron oxide is reduced by heating with carbon in a blast furnace.

Tick (✓) the box that states why aluminium oxide cannot be reduced in this way.

[1]

carbon is more reactive than aluminium	
iron is more reactive than aluminium	
aluminium is more reactive than carbon	



